



DVM S 3 Phase Water Heat Recovery  
Heating Main Target Operation

| Water Source Outdoor Unit             |              |           |                                                             |                                                                                                                |
|---------------------------------------|--------------|-----------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Component                             | Abbrevaition |           | Heating Main Mode                                           | Comments                                                                                                       |
| Compressor                            | Comp         | Operation | Target High pressure control (default: 407 psig)            | Primary function is to maintain head pressure (range 356-455psig)                                              |
|                                       |              | Range     | 20~160Hz                                                    |                                                                                                                |
| Main EEV                              | E_M          | Operation | HR - Target SH = Water temperatrue - 5.4°F                  | HP - Target SH = (Low pressure-10)*7+5                                                                         |
|                                       |              | Range     | 125 - 2000 Steps                                            | Suction Superheat 1.8°F - 12.6°F                                                                               |
| EVI EEV                               | E_EVI        | Operation | On (Open)                                                   | By-Pass Mode 18°F Superheat Between pipe in and pipe out                                                       |
|                                       |              | Range     | 0-480step                                                   | EVI Vapor Injection Mode 3.6°F Superheat Between pipe in and pipe out                                          |
| 4 Way Valve                           | V_4W         | Operation | On (Energized)                                              |                                                                                                                |
| EVI By-Pass Solenoid Valve            | V_EB         | Operation | On (Open): *T_W > 50.4 or 50Hz↓                             | By-pass mode, Discharge Temperature control                                                                    |
|                                       |              |           | Off (Close): T_W < 45°F ↓ and 65Hz↑                         | Low Ambieant heating Operation Vapor Injection circuit                                                         |
| EVI Solenoid Valve                    | V_EVI        | Operation | On (Open)                                                   |                                                                                                                |
| Hot Gas By-Pass                       | V_HG         | Operation | Off (Closed)                                                |                                                                                                                |
| Accumulator Oil Return Solenoid Valve | V_AR         | Operation | On (Open)                                                   | <b>Off</b> during start up if DSH is < 18°F. <b>Off</b> for 2 min. if DSH is < 18°F <b>On</b> when DCSH > 27°F |
| Hot Gas By-Pass 2 (HR Only)           | V_HG2        | Operation | Off Closed                                                  |                                                                                                                |
| Liquid Line Solenoid Valve            | V_L          | Operation | On (Open)                                                   |                                                                                                                |
| Main Cooling Solenoid                 | V_MC         | Operation | Off (Closed)                                                |                                                                                                                |
| Hot Gas Charging Solenoid Valve       | V_HC         | Operation | Off (Closed)                                                |                                                                                                                |
| Condenser Out Thermistor              | T_CO         | Value     | Should be 2 degrees lower than the water out temperature    | Low Side saturation temperature is " <i>typically</i> " 2-10 degrees lower than water temperature              |
| Mode Change Unit (MCU)                |              |           |                                                             |                                                                                                                |
| Component                             | Abbrevaition |           | Heating Mode                                                | Comments                                                                                                       |
| Port Heating Solenoid                 | Heat         | Operation | On (Open) Call for Heating Off (Closed) Call for Cooling    | Energized for units calling for heating                                                                        |
| Port Cooling Solenoid                 | Cool         | Operation | On (Open) Call for Cooling Off (Closed) Call for Heat       | Energized for units calling for cooling                                                                        |
| Port EEV                              | EEV          | Operation | Closed 0 Steps                                              | Open 480 step when a zone calls for cooling or during a mode change.                                           |
|                                       |              | Range     | 0 - 480 Steps                                               | Closed 0 Steps when a zone is in heating                                                                       |
| Sub_Cooling EEV                       | SC_EEV       | Operation | Active inheating Main                                       | Provide subcooling for cooling zones                                                                           |
|                                       |              | Range     | Modulating                                                  | Maintains 5.4 degrees of superheat between pipe in and out temps.                                              |
| Sub-Cooler In Temperature             |              | Operation | Should be colder then Pipe Out                              |                                                                                                                |
| Sub-Cooler Out Temperature            |              | Operation | Should be hotter then pipe in.                              |                                                                                                                |
| Indoor Unit IDU                       |              |           |                                                             |                                                                                                                |
| Component                             | Abbrevaition |           | Heating Mode                                                | Comments                                                                                                       |
| Room Temperature                      | Room Temp.   | Operation | 67-80 Degrees                                               |                                                                                                                |
| Liquid Pipe Thermistor                | EVA In 1     | Operation | Liquid Pipe Temperature for sub_cooling Control             | Used to calculate Evaporator Sub-Cooling                                                                       |
| Gas Pipe Theromistor                  | EVA Out 1    | Operation | Hot Gas Pipe Temperature should be close to discharge temp. | Used to evaluate the temperature of the hot gas coming into the coil                                           |
| EEV                                   | EEV 1        | Operation | Modulates to maintain 9 degrees of sub-cooling              | Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.                                                 |
|                                       |              | Range     | 500- 1500 Steps (80 - 120 Steps when zone is satisfied)     | Typical range between 500 and 1500 Steps                                                                       |
| IDU Fan                               | Fan Speed    | Range     | Low , Medium, High, Auto                                    |                                                                                                                |
|                                       |              |           | Low Speed: Ts - Ti  ≤ 1.8°F                                 | TS - Set Temperature                                                                                           |
|                                       |              | Operation | Medium Spreed: 1.8°F <  Ts - Ti  ≤ 5.4°F                    | Ti Indoor Temperature                                                                                          |
|                                       |              |           | High Speed 5.4°F <  Ts - Ti                                 |                                                                                                                |
| Indoor Unit IDU Cooling Mode          |              |           |                                                             |                                                                                                                |
| Component                             | Abbrevaition |           | Cooling Mode                                                | Comments                                                                                                       |
| Room Temperature                      | Room Temp.   | Operation | 67-80 Degrees                                               |                                                                                                                |
| Liquid Pipe Thermistor                | EVA In 1     | Operation | Evaporator Temperature                                      | Used to calculate Evaporator Superheat                                                                         |
| Gas Pipe Theromistor                  | EVA Out 1    | Operation | Measures Suction Pipe Temperature                           | Used to calculate Evaporator Superheat                                                                         |
| EEV                                   | EEV 1        | Operation | Modulates to maintain Superheat of 3.6 degrees F            | Superheat = to Pipe out Temp - Pipe in Temp.                                                                   |
|                                       |              | Range     | 0 - 2000 Steps (0 Steps when zone is satisfied)             | Typical range between 250 and 1500 Steps                                                                       |
| IDU Fan                               | Fan Speed    | Range     | Low , Medium, High, Auto                                    |                                                                                                                |
|                                       |              |           | Low Speed: Ts - Ti  ≤ 1.8°F                                 | TS - Set Temperature                                                                                           |
|                                       |              | Operation | Medium Spreed: 1.8°F <  Ts - Ti  ≤ 5.4°F                    | Ti Indoor Temperature                                                                                          |
|                                       |              |           | High Speed 5.4°F <  Ts - Ti                                 |                                                                                                                |